

Batched Dense Optical Flow in PyTorch

批量稠密光流的 PyTorch 实现与基准测试

TV-L1 & DeepFlow: OpenCV CPU vs. CUDA vs. pure-torch reimplementation

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Middlebury pairs 480×640 · NVIDIA L40 (SLURM, ilps-cn119) · 48-core CPU login node

- ▶ Dense optical flow is a standard **preprocessing step at scale**: video understanding, motion features, temporal augmentation, dataset curation — millions of frame pairs, not one.
- ▶ Classic accurate methods (TV-L1, DeepFlow) are **variational**: hundreds of small iterations per pair $\Rightarrow$ slow on CPU, awkward on GPU.
- ▶ What matters for preprocessing is **batch throughput** (pairs/second over a whole dataset), not single-pair latency.
- ▶ Goal: benchmark OpenCV CPU vs. OpenCV CUDA vs. our **batched pure-PyTorch reimplementations**, on identical inputs and parameters.

中文要点 稠密光流常作为大规模视频处理的预处理步骤：要处理的是数百万帧对，而非单帧。经典变分方法（TV-L1、DeepFlow）迭代次数多、单对速度慢；预处理场景真正重要的是批量吞吐量。我们在相同输入与参数下对比 OpenCV CPU / OpenCV CUDA / 自研批量化 PyTorch 实现。

Data & hardware

- ▶ Middlebury training pairs, 480×640 grayscale (also downscaled 240×320); ground-truth flow for EPE.
- ▶ GPU: NVIDIA L40 (142 SMs), SLURM node `ilps-cn119`.
- ▶ CPU: 48-core login node.
- ▶ Env: `uv-managed`; `opencv-contrib-python-headless`, PyTorch.

Algorithm × backend matrix

	cv CPU	cv CUDA	torch
TV-L1	✓	✓*	✓
DeepFlow	✓	none	✓
Farneback	✓	✓*	—

*only with a custom CUDA build of OpenCV — pip wheels ship **without** CUDA.

中文要点 数据：Middlebury 训练集（ 480×640 ，含真值光流，用于计算端点误差 EPE）。硬件：NVIDIA L40（142 个 SM，SLURM 节点）对比 48 核 CPU 登录节点。注意：pip 安装的 OpenCV 不带 CUDA，需要自行编译；DeepFlow 在 OpenCV 中没有任何 CUDA 版本。

$$E(\mathbf{u}) = \int_{\Omega} \lambda |\rho(\mathbf{u}(\mathbf{x}))| + |\nabla u_1| + |\nabla u_2| \, d\mathbf{x}$$

with the image residual linearized around the current warp $\mathbf{u}^0$:

$$\rho(\mathbf{u}) = l_1(\mathbf{x} + \mathbf{u}^0) + \langle \nabla l_1(\mathbf{x} + \mathbf{u}^0), \mathbf{u} - \mathbf{u}^0 \rangle - l_0(\mathbf{x})$$

- ▶ L^1 data term: robust to outliers/occlusions; TV term: allows sharp flow discontinuities at motion boundaries.
- ▶ Both terms non-smooth $\Rightarrow$ solved via **duality**, not gradient descent.
- ▶ Params used (OpenCV defaults): $\lambda = 0.15$, $\theta = 0.3$, $\tau = 0.25$, 5 scales (step 0.8), 5 warps, up to 10×30 primal-dual iterations.

中文要点 能量 = 全变差正则项 + 鲁棒 L^1 数据项； ρ 是在当前解 $\mathbf{u}^0$ 处线性化的亮度残差。 L^1 对遮挡/离群点鲁棒，TV 项保留运动边界的不连续性。两项都不可微，因此用对偶方法求解而非梯度下降。

TV-L1: Duality Splitting & Iterations 对偶分裂与迭代

Auxiliary field $\mathbf{v}$ decouples the two terms (θ -coupling):

$$\min_{\mathbf{u}, \mathbf{v}} \int_{\Omega} |\nabla u_1| + |\nabla u_2| + \frac{1}{2\theta} \|\mathbf{u} - \mathbf{v}\|^2 + \lambda |\rho(\mathbf{v})| \, d\mathbf{x}$$

1) **Pointwise thresholding** (closed form in $\mathbf{v}$):

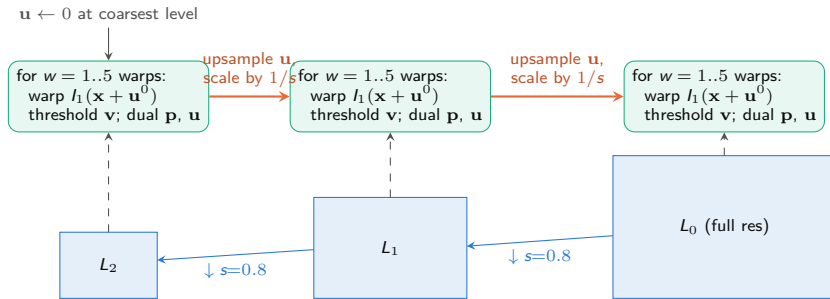
$$\mathbf{v} = \mathbf{u} + \begin{cases} \lambda\theta \nabla I_1 & \text{if } \rho(\mathbf{u}) < -\lambda\theta |\nabla I_1|^2 \\ -\lambda\theta \nabla I_1 & \text{if } \rho(\mathbf{u}) > \lambda\theta |\nabla I_1|^2 \\ -\rho(\mathbf{u}) \nabla I_1 / |\nabla I_1|^2 & \text{if } |\rho(\mathbf{u})| \leq \lambda\theta |\nabla I_1|^2 \end{cases}$$

2) **TV proximal step** via dual variables $\mathbf{p}_d$ (Chambolle), for $d = 1, 2$:

$$u_d = v_d + \theta \operatorname{div} \mathbf{p}_d, \quad \mathbf{p}_d^{n+1} = \frac{\mathbf{p}_d^n + \frac{\tau}{\theta} \nabla u_d^{n+1}}{1 + \frac{\tau}{\theta} |\nabla u_d^{n+1}|} \quad (\text{projection onto } |\mathbf{p}_d| \leq 1), \quad \tau \leq 1/4$$

中文要点 引入辅助变量 $\mathbf{v}$ 将两项解耦：第一步对每个像素做闭式阈值化 更新 $\mathbf{v}$ ；第二步用对偶变量 $\mathbf{p}$ 做 TV 近端步（分母即到单位球的投影）。两步都是纯逐像素运算——非常适合张量化。

Coarse-to-Fine Pyramid + Warping 由粗到细金字塔与图像变形



Linearization ρ is only valid for *small* displacements $\Rightarrow$ estimate coarse motion on small images, then warp I_1 by the upsampled flow and re-linearize at each finer level.

中文要点 线性化残差 ρ 只对小位移成立：先在低分辨率估计大运动，再逐层上采样光流（数值乘 $1/s$ ），用它对 I_1 做 warp 后重新线性化。每层做 5 次 warp，每次 warp 内做数百次原始-对偶迭代。

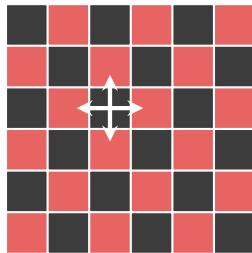
$$E(\mathbf{w}) = \int_{\Omega} \left[\begin{aligned} &\delta \Psi(|I_2(\mathbf{x} + \mathbf{w}) - I_1(\mathbf{x})|^2) && \text{brightness constancy} \\ &+ \gamma \Psi(\|\nabla I_2(\mathbf{x} + \mathbf{w}) - \nabla I_1(\mathbf{x})\|^2) && \text{gradient constancy} \\ &+ \alpha \Psi(\|\nabla u\|^2 + \|\nabla v\|^2) \end{aligned} \right] d\mathbf{x} && \text{smoothness}$$

- ▶ Full DeepFlow adds a matching term βE_{match} driving the flow toward **DeepMatching** correspondences.
- ▶ **OpenCV's** `createOptFlow_DeepFlow` implements only the variational part — no DeepMatching ($\beta = 0$).
- ▶ Params (OpenCV defaults): $\alpha = 1$, $\delta = 0.5$, $\gamma = 5$, pyramid downscale 0.95 ($\sim 40+$ levels), 5 fixed-point iterations, 25 SOR iterations, $\omega = 1.6$.

中文要点 能量含亮度守恒 (权重 δ)、梯度守恒 (γ) 与平滑项 (α)，均用鲁棒罚函数 $\Psi(s^2) = \sqrt{s^2 + \varepsilon^2}$ 。完整 DeepFlow 另有 DeepMatching 匹配项；OpenCV 只实现了变分部分 (无匹配项)。

DeepFlow: Fixed-Point + Red-Black SOR 不动点迭代与红黑 SOR

- ▶ Euler–Lagrange equations are nonlinear in $\mathbf{w} = (u, v)$: **fixed-point linearization** — freeze Ψ' weights and image derivatives at $\mathbf{w}^k$, solve for the increment $d\mathbf{w}$.
- ▶ Each linearization yields a large **sparse linear system** (5-point stencil per pixel, 2 coupled channels).
- ▶ Solved by **SOR** ($\omega = 1.6$): each pixel update mixes its own residual with already-updated neighbors.
- ▶ **Red-black ordering**: a red pixel's stencil touches only black pixels, so *all red pixels update simultaneously*, then all black — two fully parallel half-sweeps.



A red update reads only **black** neighbors $\Rightarrow$ each half-sweep is one *vectorized masked update*.

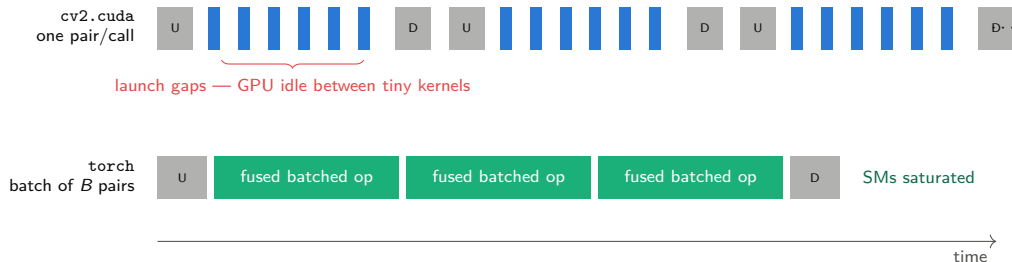
中文要点 外层不动点迭代固定 Ψ' 权重得到稀疏线性方程组，内层用 SOR ($\omega = 1.6$) 求解；红黑排序下红格只依赖黑格，红/黑各自可全并行更新——即两次带掩码的张量化更新。

Why OpenCV GPU Is Limited Here OpenCV GPU 路线的局限

1. **Packaging:** pip opencv-python wheels ship *without* CUDA $\Rightarrow$ custom source build per cluster/CUDA version — a real operational burden.
2. **API is one-pair-at-a-time:** cv2.cuda takes a single GpuMat pair. A 480×640 (let alone 240×320) problem cannot fill an L40's **142 SMs**; upload/download + kernel-launch overhead dominates.
3. **Hundreds of tiny sequential iterations:** TV-L1 ≈ 5 scales $\times 5$ warps $\times$ up to 300 primal-dual steps $\Rightarrow$ thousands of small kernels per pair — *launch-bound*, GPU mostly idle.
4. **DeepFlow: no CUDA version exists in OpenCV at all**; and a naive (lexicographic) SOR sweep is inherently sequential — it must be re-ordered (red-black) to parallelize.

中文要点 (1) pip 版 OpenCV 不含 CUDA, 须自行编译; (2) cv2.cuda 一次只处理 一对图像, 小图无法喂饱 L40 的 142 个 SM, 上传/下载与核启动开销占主导; (3) TV-L1 有成千上万个微小串行核, 属于启动开销受限; (4) DeepFlow 根本没有 OpenCV CUDA 版本, 且朴素 SOR 天生串行。

Launch-Bound vs. Batched 核启动受限 vs. 批量化



Same total math — but one launch now covers B images' worth of pixels, so arithmetic hides the launch latency and fills the GPU.

中文要点 上：cv2.cuda 每对图像触发一串微小核函数，核间空隙里 GPU 空转。下：批量张量把 B 对图像合成一次大核启动——总计算量相同，但每次启动的工作量足以摊薄启动延迟并填满所有 SM。

Our Approach: Pure-Torch, Batched 纯 PyTorch 批量化实现

`torch_flow.py (calc_flow_tv11)` · `torch_deepflow.py (calc_flow_deepflow)`

- ▶ **Batched end to end:** (B, H, W) image pairs $\rightarrow (B, 2, H, W)$ flows; every step is a vectorized tensor op over the whole batch — no Python loop over pairs.
- ▶ **Warping** = `grid_sample` (bilinear, one call for the batch).
- ▶ **Red-black SOR / primal-dual** steps as masked tensor updates: two half-sweeps = two fused elementwise ops.
- ▶ **Same code on CPU and CUDA** — just `.to(device)`; no custom OpenCV build.
- ▶ **Composable** with a torch pipeline: `DataLoader` feeding, `torch.no_grad()` inference, `torch.compile` for kernel fusion.

中文要点 输入 (B, H, W) 帧对、输出 $(B, 2, H, W)$ 光流，全流程张量化：warp 用 `grid_sample`，红黑 SOR / 原始-对偶迭代化为带掩码的融合逐元素运算。同一份代码在 CPU 与 GPU 上直接运行，且天然融入 torch 数据管线（`DataLoader`、`no_grad` 推理、`torch.compile`）。

Runtime Results 运行时间结果

Per-pair wall time at 240×320 (Middlebury), identical algorithm parameters across backends.

Algorithm	cv CPU	cv CUDA	torch CPU	torch CUDA (L40)
TV-L1	424 ms (2.4 FPS)	21.1 ms ^a	8300 ms	5.5 ms^b
DeepFlow	346 ms (2.9 FPS)	— (none)	2263 ms	16.0 ms^c (fp16: 7.4)
Farneback	84 ms @ 480×640	—	—	—

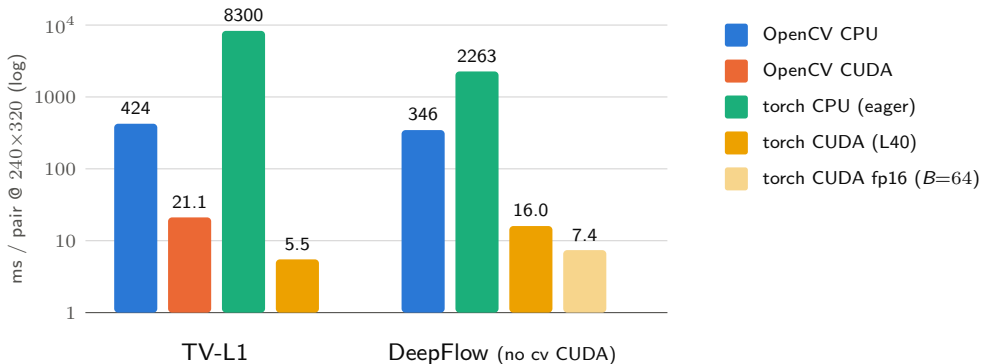
^asingle pair, GPU-resident; 9.3 ms with 4 streams $\times$ 4 solver instances; needs custom CUDA 12.9 OpenCV build.

^btorch.compile + adaptive early exit, $B=16$. ^ctorch.compile, $B=64$; per pair = batch time / B .

- ▶ Multicore doesn't help OpenCV here (Farneback 1 vs 48 threads: 85 vs 84 ms); torch eager CPU is even slower — as expected, per-op overhead without fusion.
- ▶ Batched torch CUDA: **76 $\times$** over OpenCV CPU (TV-L1) and **1.7 $\times$ faster than cv2.cuda's 4-stream ceiling** while computing the *full* algorithm.

中文要点 最终结果：TV-L1 torch CUDA 每对 5.5 ms (compile + 自适应提前退出, $B=16$)，比 OpenCV CPU 快约 76 倍，比 cv2.cuda 四流并发上限 (9.3 ms) 还快 1.7 倍；DeepFlow 16.0 ms (fp16 为 7.4 ms)。torch eager CPU 反而更慢——符合预期，逐算子开销大且无融合。

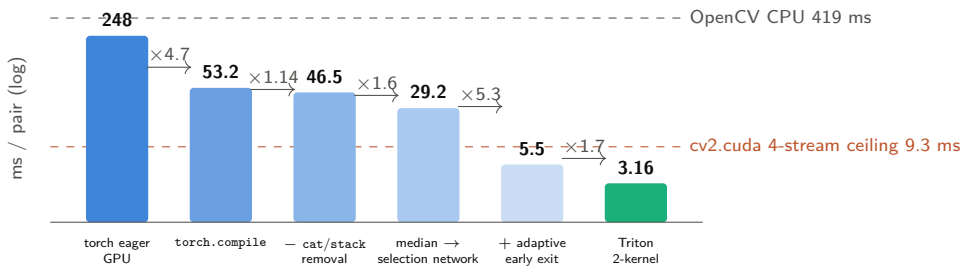
Runtime Chart 运行时间对比图



Log scale — each gridline is 10 $\times$. torch CUDA spans 3 orders of magnitude below torch eager CPU.

中文要点 对数坐标（每格 10 倍）：torch CUDA（黄）在两种算法上都比 OpenCV CPU 低约两个数量级；TV-L1 上也低于 cv2.cuda（橙）。torch eager CPU（绿）最慢，印证瓶颈在逐算子开销而非算力。

Optimization Waterfall: TV-L1 on L40 优化阶梯



240×320, $B=16$: **16.8×** over plain compile, **78×** over eager, **133×** over OpenCV CPU — the cv2.cuda 4-stream line is now **crossed**.

- ▶ Median filter was 47% of GPU time → replaced by a 202-op min/max **selection network** (18× on that kernel).
- ▶ Early exit = OpenCV's own throttled-sync scheme — *bit-for-bit result-neutral*. (CUDA graphs variant: $\sim 2\times$ slower than compile, per-step clones.)
- ▶ Final step: **2 hand-written Triton kernels** for the inner step — exactly **2.00 launches/iteration** vs Inductor's 6.00; median filtering *kept*; EPE vs cv2 0.0263 (compile's noise floor).

中文要点 eager 248 → compile 53.2 → 去 cat/stack 46.5 → 选择网络 29.2 → 提前退出 5.5 → 手写 Triton 双核 3.16 ms (每迭代恰 2 次核启动, Inductor 为 6 次; 保留中值滤波)。合计快 OpenCV CPU 133 倍, 已越过 cv2.cuda 四流上限。

How Strong Is the cv2.cuda Baseline? cv2.cuda 基线有多强

- ▶ Single pair, resident: only **42–47% utilization** (launch-latency-bound); 4 streams $\times$ 4 instances give **2.3 $\times$** , **not 4 $\times$** (92% util).
- ▶ Their CUDA TV-L1 is a **lighter variant** (per OpenCV's own parity test): *no median filter*, real early exit, bicubic $a=-0.5$ vs $-0.75 \Rightarrow$ **2.5–3 $\times$ fewer DRAM bytes per iteration**.
- ▶ **DeepFlow**: no GPU version exists (`cudaoptflow = {brox, farneback, nvof, pyrllk, tvl1}`); our 16.0 ms (fp16: 7.4) vs their CPU 297 ms = **18.6 $\times$** (fp16: **40 $\times$**).

	Resolution	cv2.cuda (variant)		ours (full algo)	
		1 pair	4-stream	compile	Triton
TV-L1 head-to-head, ms/pair (fp32):	240 $\times$ 320	21.1	9.3	5.5	3.16
	480 $\times$ 640	26.5	13.1	22.6	14.1
	480 $\times$ 854	27.2	14.0	27.3	15.5
	720 $\times$ 1280	35.8	21.6	62.0	33.2

*Verdict: hand-fusion was the remaining gap, and porting OpenCV's own 2-kernel split to Triton closes it — 2.9 $\times$ past their 4-stream ceiling at 240p, at parity with it at 480p, and at 720p we now come in **under their single-pair figure** while doing the median filtering they skip. Only their 4-stream ensemble stays ahead (1.54 $\times$).*

中文要点 cv2.cuda 单对利用率仅 42--47%，四流上限 2.3 倍；其 CUDA 版为简化变体。结论：此前唯一的差距在手写核融合，把 OpenCV 的双核拆分移植到 Triton 后已被补上——240p 超其四流上限 2.9 倍，480p 与之持平，720p 更是在 保留其省略的中值滤波前提下低于其单对成绩；仅其四流并发仍领先 1.54 倍。

Scaling with Resolution & GPU Utilization 分辨率扩展与利用率

ms/pair, adaptive early exit (TV-L1), best batch size per resolution.

Resolution	TV-L1				DeepFlow		
	B	compile	Triton ^a	fp16 ^b	B	fp32	fp16
240×320	16	5.5	3.16	2.17	64	16.0	7.4
480×640	4	22.6	14.1	15.2	16	71.7	39.5
480×854	4	27.3	15.5	19.0	16	99.4	49.1
720×1280	2	62.0	33.2	41.9	8	236.7	— ^c

^anew best: 2 Triton kernels, fp32 + early exit, 1.34–2.0× over compile everywhere. ^bfp16 (compile) at doubled batch — half the working set, so the optimal B doubles. ^cDeepFlow fp16 **overflows (NaN)** at 720p — use fp32 there.

- ▶ Saturation batch size falls $\sim 1/\text{pixels}$: once the GPU is full, bigger B stops helping.
- ▶ **L2-residency cliff**: working set $\approx 3.7\text{ MB} \times B$ vs 96 MB L2 predicts the optima ($B=16$ @240p, $B=4$ @VGA) *exactly* — and fp16's halved working set doubles them.
- ▶ GPU util 67–92% across the sweep; high “utilization” $\neq$ efficient (SMs counted busy while waiting on DRAM).

中文要点 Triton 列为最新最佳（各分辨率较 compile 快 1.34--2.0 倍）。最优批大小随像素数近似反比下降；L2 驻留悬崖：工作集 $\approx 3.7\text{ MB} \times B$ 对 96 MB L2，恰好预测出 240p 的 $B=16$ 与 VGA 的 $B=4$ ；fp16 工作集减半，最优 B 翻倍。DeepFlow fp16 在 720p 溢出（NaN），须用 fp32。

- ▶ **fp16 > bf16 for iterative solvers** — mantissa matters (11 vs 8 bits): DeepFlow bf16 +74% EPE; fp16 only +0.003 px.
- ▶ fp16 pays off **only at large B** (bandwidth-bound regime): TV-L1 **1.75 $\times$** , DeepFlow **2.10 $\times$** at $B=64$; at small B (launch/latency-bound) it buys nothing.
- ▶ **TF32 / autocast are no-ops** here — these solvers contain no matmuls; everything is elementwise/stencil.
- ▶ **Keep the 0..255 intensity scale**: rescaling images to 0..1 silently breaks both solvers' tuned thresholds (TV-L1 EPE 0.21 $\rightarrow$ 1.46).

中文要点 迭代求解器更吃尾数位：bf16（8位尾数）使 DeepFlow EPE 恶化 74%，而 fp16（11位）仅差 0.003 px。fp16 只在大批量（带宽受限区）才有收益（ $B=64$ ：TV-L1 1.75 倍、DeepFlow 2.10 倍）；TF32/autocast 无效（没有矩阵乘）。务必保持 0..255 灰度尺度——归一化到 0..1 会破坏两个算法的阈值（TV-L1 EPE 从 0.21 恶化到 1.46）。

Accuracy: Endpoint Error 精度：端点误差

Mean EPE on Middlebury pairs at 240×320 (pixels with GT magnitude $> 10^9$, i.e. “unknown”, excluded).

Algorithm	Backend	EPE vs. GT	EPE vs. OpenCV ref.
TV-L1	OpenCV CPU	0.208	(reference)
TV-L1	torch	0.208	0.027
DeepFlow	OpenCV CPU	0.186	(reference)
DeepFlow	torch	0.186	0.000*

* $\approx 10^{-6}$ px — the DeepFlow port is **bit-identical** to OpenCV.

- ▶ **Full fidelity:** torch matches OpenCV's own GT accuracy exactly (TV-L1: 0.208 both); the 0.027 px TV-L1 gap vs the reference comes from interpolation/boundary choices only.
- ▶ DeepFlow's variational part remains the more accurate algorithm (0.186 vs 0.208 EPE) — and our port reproduces it to the bit.

中文要点 精度结论：torch 版与 OpenCV 对真值的 EPE 完全一致（TV-L1 双方均 0.208）；与 OpenCV 参考流场相比，TV-L1 差 0.027 px（仅插值/边界差异），DeepFlow 则逐位一致（差约 10^{-6} px）。DeepFlow（0.186）仍比 TV-L1（0.208）更准。

- ▶ Video decode $\rightarrow$ chunked batches $\rightarrow$ flow on GPU: **transfers are** $< 2\%$ of end-to-end runtime — compute-bound, as designed.
- ▶ Best chunk size: **16 frames (TV-L1)**, **32 frames (DeepFlow)**.
- ▶ Sustained throughput at 240×320 on *one* L40: **~ 180 FPS TV-L1** (with early exit), **~ 67 FPS DeepFlow** (fp16: **135 FPS**).
- ▶ Footprint < 1.7 GB $\Rightarrow$ multiple video streams can share a single GPU.

中文要点 视频解码 $\rightarrow$ 分块成批 $\rightarrow$ GPU 光流：数据搬运占端到端时间不到 2%。最优块长 TV-L1 为 16 帧、DeepFlow 为 32 帧。单张 L40 在 240×320 下约 180 FPS (TV-L1，含提前退出)、67 FPS (DeepFlow；fp16 达 135 FPS)，显存占用低于 1.7 GB，可多路视频共享一卡。

Conclusions

- ▶ Batching + `torch.compile` + **algorithm-level porting** (median $\rightarrow$ selection network, adaptive early exit) beats a hand-written CUDA baseline at *full* fidelity (DeepFlow bit-identical).
- ▶ OpenCV-CUDA's tricks **transfer cleanly to torch**: fusion via compile, `calcError`-style split, adaptive/throttled sync.
- ▶ When compile's ~ 6 kernels/iteration still isn't enough, a **2-kernel Triton port of the same split** recovers hand-CUDA launch efficiency *without leaving Python*: one file, still batched, deterministic reduction for bitwise batch consistency.
- ▶ Red-black reordering makes SOR tensor-friendly; the L2-residency rule picks the batch size per resolution.

Practical notes

- ▶ uv project; `opencv-contrib-python-headless` on compute nodes; login node has no GPU — the benchmark script auto-skips unavailable backends.
- ▶ GPU runs via SLURM: `srun -p gpu -w ilps-cn119 --gres=gpu:1 --cpus-per-task=8 --mem=16G uv run --no-sync python benchmark.py`

中文要点 批量化 + compile + 算法级移植在完整保真度下已胜过手写 CUDA 基线；当 compile 每迭代约 6 次核启动仍不够时，用 Triton 重写同一双核拆分（2 次启动）即可不离开 Python 取得手写 CUDA 的启动效率，并保持批量化与逐位一致。

1. C. Zach, T. Pock, H. Bischof. *A Duality Based Approach for Realtime TV-L1 Optical Flow*. DAGM Symposium on Pattern Recognition, 2007.
2. P. Weinzaepfel, J. Revaud, Z. Harchaoui, C. Schmid. *DeepFlow: Large Displacement Optical Flow with Deep Matching*. ICCV 2013.
3. J. Sánchez Pérez, E. Meinhardt-Llopis, G. Facciolo. *TV-L1 Optical Flow Estimation*. Image Processing On Line (IPOL), vol. 3, 2013.
4. S. Baker, D. Scharstein, J.P. Lewis, S. Roth, M.J. Black, R. Szeliski. *A Database and Evaluation Methodology for Optical Flow*. IJCV, 2011. (Middlebury benchmark)
5. OpenCV documentation: `cv::optflow::DualTVL1OpticalFlow`,
`cv::optflow::createOptFlow_DeepFlow`, `cv::cuda::OpticalFlowDual_TVL1`.

中文要点 以上为 TV-L1 (Zach 等 2007 ; Sánchez 等 IPOL 2013 实现)、DeepFlow (Weinzaepfel 等 ICCV 2013)、Middlebury 数据集及 OpenCV 文档的出处。